

Momentum Equations

- Linear * Inertial Ref. Frames
- * Impulse
- * Change of Basis
- Angular * Inertial Ref. Frames

HW3 due Fri 6pm

→ 3 questions
↳ Gradescope→ 1 question
↳ Matlab grader.

Linear Momentum

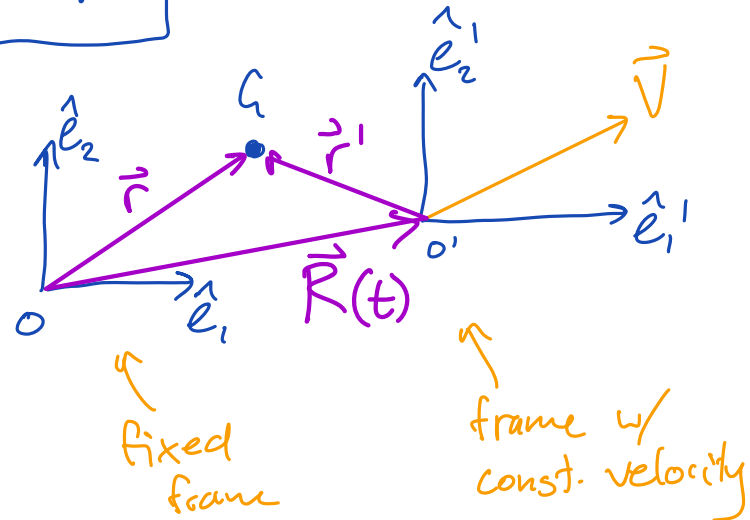
$$\sum \vec{F}_{ext} = \dot{\vec{p}}$$

I $\neq$

* Inertial Reference Frames

$$\Rightarrow \vec{F}_a = m \vec{a}_{oa}$$

acceleration of
a w.r.t. I Ref. frame.



$$\Rightarrow \vec{r} = \vec{R} + \vec{r}'$$

$$\vec{r}' = -\vec{R} + \vec{r}$$

where $\vec{R}(t) = \vec{R}_0 + \vec{V}t$

$$\frac{d}{dt} \Rightarrow \frac{d\vec{r}'}{dt} = -\frac{d\vec{R}}{dt} + \frac{d\vec{r}}{dt} \Rightarrow \vec{v}' = -\vec{V} + \vec{v}$$

$$\frac{d}{dt} \Rightarrow \frac{d\vec{v}'}{dt} = -\frac{d\vec{V}}{dt} + \frac{d\vec{v}}{dt} \Rightarrow \vec{a}' = \vec{a}$$

After class
correction.

The '+' and '-'
were reversed
in these lines.

The result was
accurate.

if $\vec{a} = \vec{a}' = 0$, then $\sum F = 0$, $\dot{\vec{p}} = 0 \Rightarrow \vec{p} = \text{const.}$

$\therefore$ Conservation of Linear Momentum holds in
any I Ref. Frame.

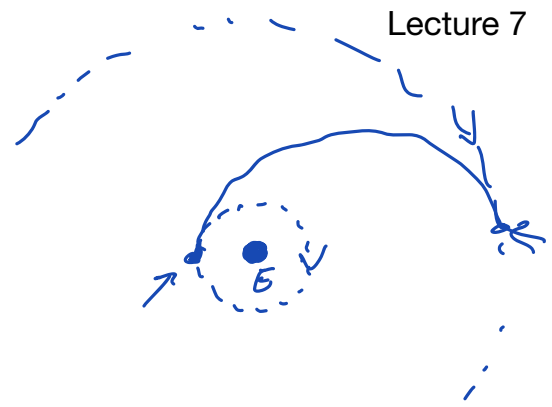
* Impulse

Take integral of $\vec{F} = m\vec{a}$

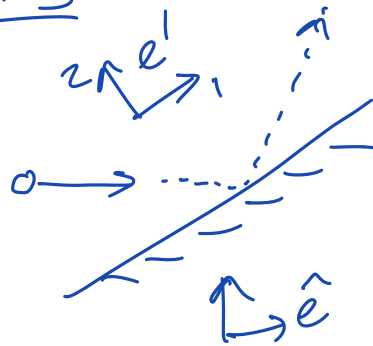
$$\Rightarrow \int \vec{F} dt = \int m\vec{a} dt.$$

$$\int_{t_1}^{t_2} \vec{F} dt = \int_{t_1}^{t_2} m\vec{a} dt = m(\vec{v}_2 - \vec{v}_1) = m\Delta \vec{v}$$

$$\int_{t_1}^{t_2} \vec{F} dt = \Delta \vec{p} \Rightarrow \text{impulse / change in lin. momentum over some time.}$$



* Change of Basis



$$\dot{\vec{p}} \cdot \hat{e}_1 = 0$$

$$\vec{p} \cdot \hat{e}_1 = \text{const.}$$

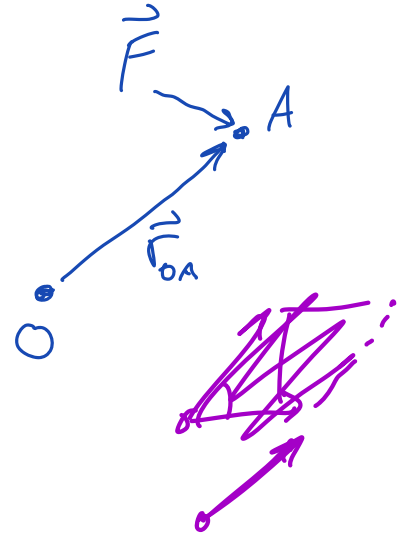
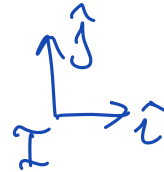
Angular Momentum

$$\sum \vec{M}_{ext/O} = \dot{\vec{H}}_{/O}$$

RoC
angular momentum

* Inertial Reference Frame

'O' is fixed in Frame.



$$\vec{r}_{OA} \times \vec{p}_A = \vec{r}_{OA} \times m \vec{v}_A$$

$$\frac{d}{dt} \Rightarrow \frac{d}{dt}(\vec{r}_{OA}) \times \vec{p}_A + \vec{r}_{OA} \times \frac{d}{dt}(\vec{p}_A) = m \left[\frac{d}{dt}(\vec{r}_{OA}) \times \vec{v}_A + \vec{r}_{OA} \times \frac{d}{dt}(\vec{v}_A) \right]$$

$$\cancel{\vec{v}_A \times \vec{p}_A} + \vec{r}_{OA} \times \dot{\vec{p}}_A = m \left[\cancel{\vec{v}_A \times \vec{v}_A} + \vec{r}_{OA} \times \dot{\vec{v}}_A \right]$$

$\underbrace{\quad}_{\vec{F}}$
 $M_{A/O} = \dot{\vec{H}}_{A/O}$

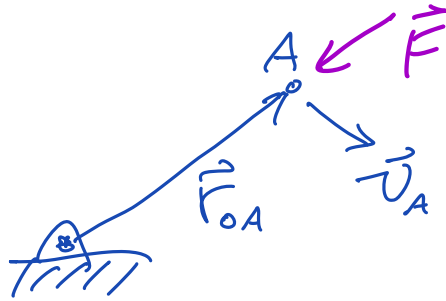
• Different fixed point in I will produce diff. $\vec{H}$,

$$\sum \vec{M}_O = 0, \dot{\vec{H}}_O = 0, \vec{H} = \text{const.}$$

$\Rightarrow$ The Conservation of Angular Momentum does not hold if 'O' is not fixed.

*Central Force Problems (HW3)

inc. orbits, electrostatic forces



Force directed
towards center.

$$\vec{r}_{OA} \times \vec{F} = 0$$

$$\therefore H_o = \text{const.}$$

HW/

a) Find $\vec{F}(\vec{r})$ such that v is independent of $\vec{r}$.

b) Find and Plot a periodic trajectory that is not circular.

